

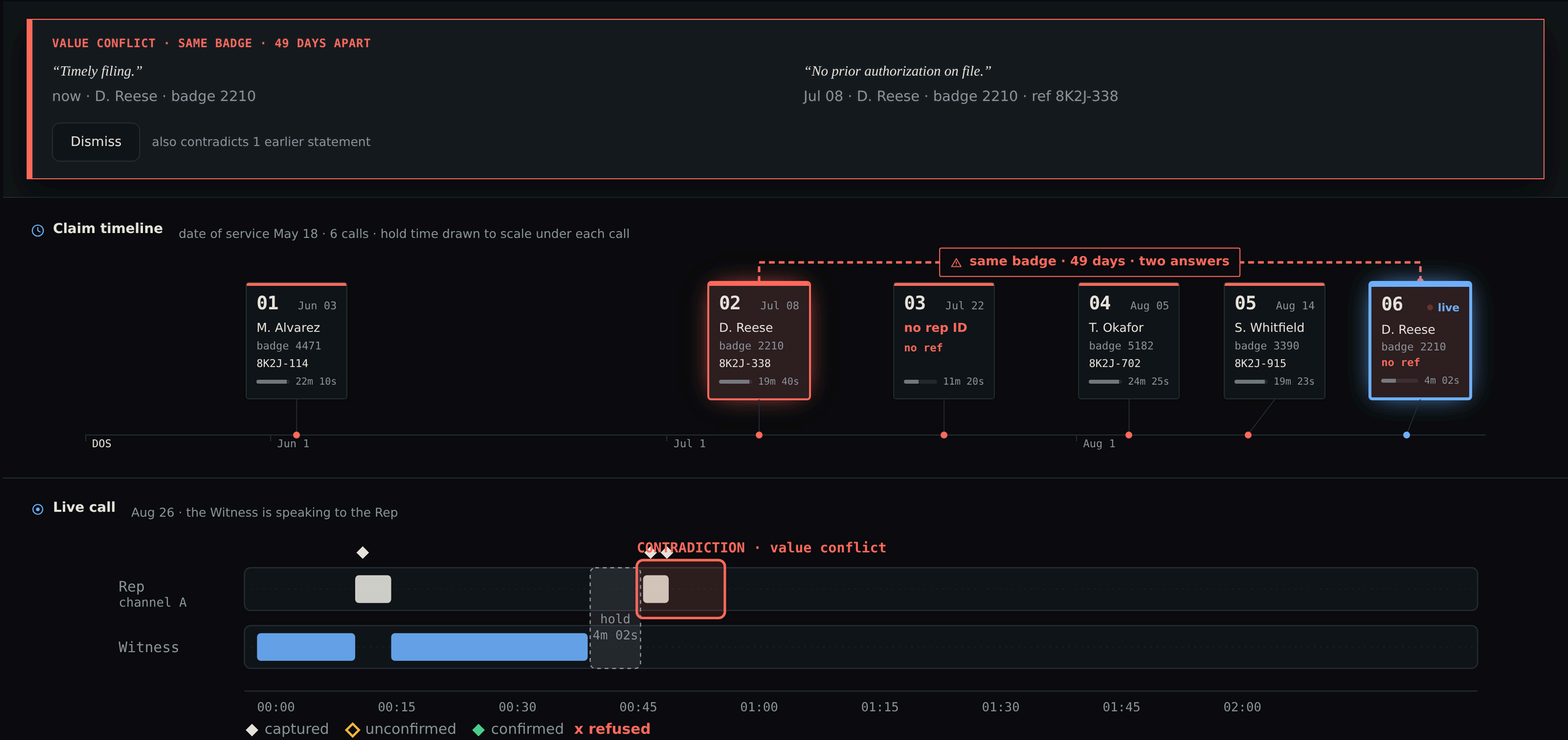
The Witness

Payer call memory

We don't predict what the payer will pay.
We record what the payer said.

Same representative, badge 2210.
49 days apart. Two answers.
Caught on the recorded line, in real time.

AssemblyAI Streaming STT v3 + Voice Agent API
AssemblyAI Voice Agent Hackathon, lablab.ai, September 2026
the-witness-omega.vercel.app



🎙 Live call

Aug 26 · the Witness is speaking to the Rep

Rep channel A

hold 4m 02s

Witness

CONTRADICTION · value conflict

00:00

00:15

00:30

00:45

01:00

01:15

01:30

01:45

02:00

◆ captured

◆ unconfirmed

◆ confirmed

✗ refused

The answer is spoken, and then it is gone

"It's denying on a diagnosis code."

"Which one?"

"I don't have that level of detail on my end."

25 min

and \$13.80 of staff time per manual phone claim-status inquiry

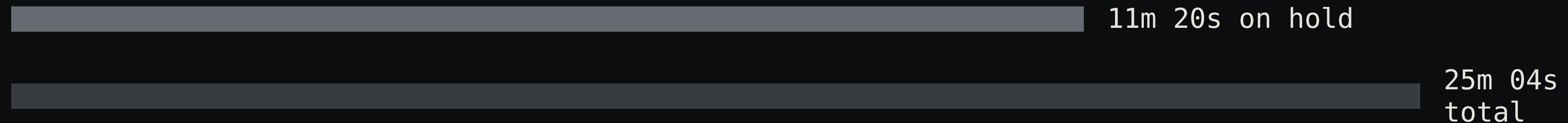
19%

of in-network claims denied; under 1% of denials ever appealed

Weeks later a different representative gives a different answer, and nothing in the revenue cycle notices. Appealing means reconstructing what the payer said, and nobody can.

What vanishes: one call from the demo claim

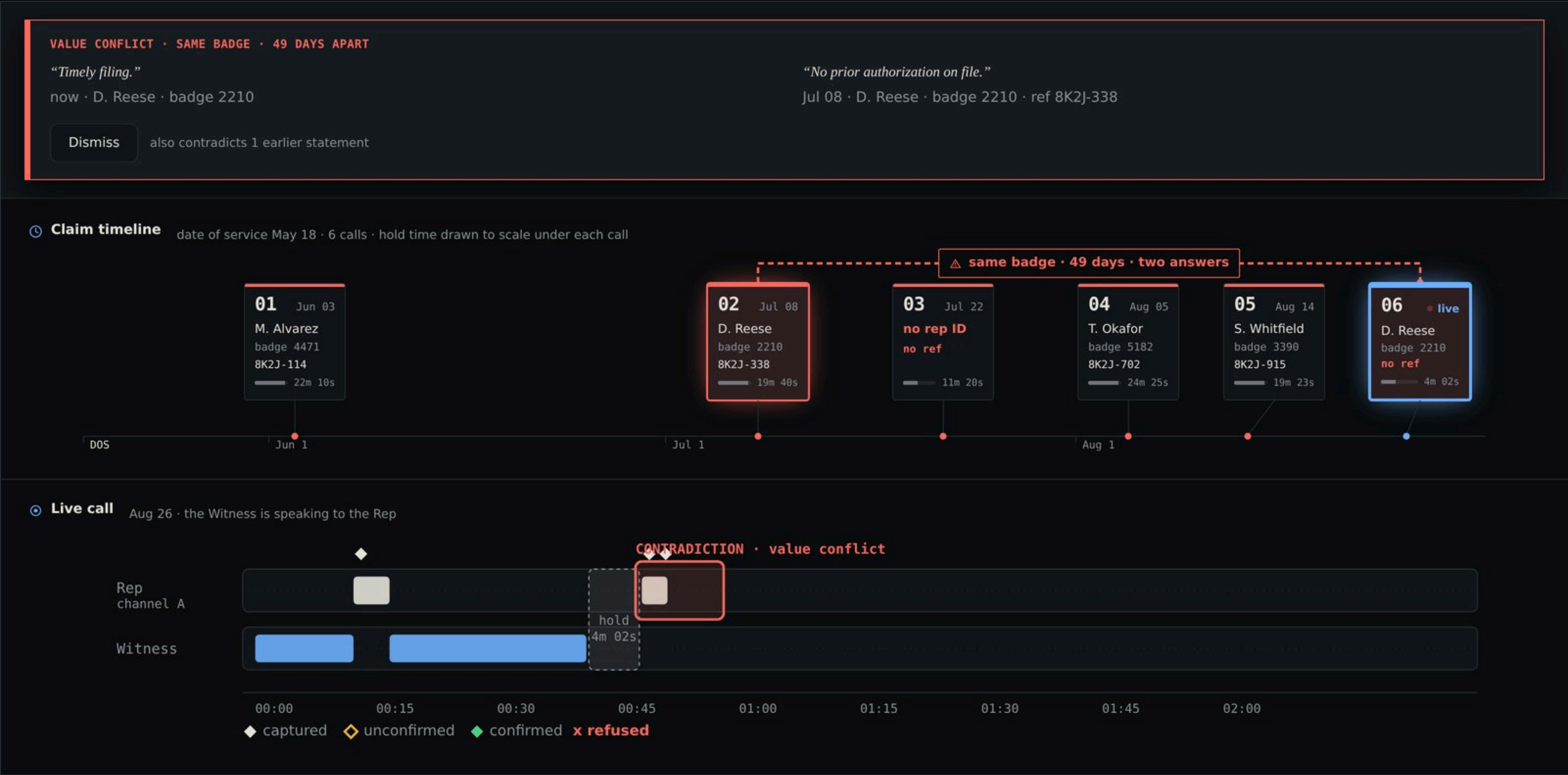
Call 03, July 22. Twenty-five minutes on the phone. The rep would not give a name or badge, would not say which diagnosis code, would not give a reference number.



The claim note afterwards reads "denied, dx issue, call back". Everything that could be quoted in an appeal is gone. Three refusals happened and none of them is on paper.

The Witness records a refusal as a row: REFUSAL · asked which diagnosis code · rep declined to specify · 14:37. The absence is the evidence.

The product in one frame



How it works: two sockets, one memory

The Rep's audio goes to Streaming STT v3. Transcription only, verbatim, never cleaned. Keyterms are seeded from the claim and the ledger before the call.

The Witness speaks through the Voice Agent API. It is never sent audio. It never calls tools. It never writes the ledger. By choice.

Speaker identity is structural: which socket carried the audio. Never diarization.

Every line the Witness speaks is assembled by our code from statement rows and handed over with `reply.create`. A self-check compares every number it actually says against the record.

The model is a mouth, not a hand. A sentence the model invented cannot be played back, and playing it back is the whole product.

The evidence model

Append-only ledger. A statement is never updated or deleted. A later, conflicting statement is a new row. The history is the product.

Hash-chained. Each row's hash covers the one before it. The appeal packet carries the chain head; anyone holding the log can recompute it.

Five contradiction kinds plus refusals. Value conflict, existence denial, status flip, commitment violation, policy inconsistency. A refusal is a row, not a blank.

Confidence is a state, never a percentage: confirmed, unconfirmed, not captured. An unconfirmed reference number triggers a spoken read-back.

The before-you-hang-up gate is an exit condition: the call does not close while a required field is missing and not refused.

Why AssemblyAI, specifically

Spoken alphanumerics are the hard case in speech recognition. Reference numbers, member IDs and badge numbers are exactly that, and a wrong one is a fabricated fact in an appeal.

- **Keyterms seeded before the call** with the claim's own identifiers: open-vocabulary transcription becomes candidate matching.
- **Read-back, not a guess.** *"Let me read that back. Eight K two J, nine eight eight. Did I get that right?"*
- **Verbatim on purpose.** The Dictation API was refused for the Rep channel: it resolves "timely fil-, sorry, no prior auth" into a clean sentence and discards the contradiction.
- **Turn detection tuned for identifiers** so a reference read with a pause stays one turn.

No accuracy percentage is claimed anywhere. Degraded audio is the test input, and the read-back loop is what is shown handling it.

What is live, what is replayed

Layer	State
Statement extraction, contradiction engine, ledger, hash chain, call plan, speech assembly	built, 205 unit tests
Console, appeal packet, worklist import, multi-patient roster	built, deployed
Live Mode A on two AssemblyAI sockets, in-browser call recorder	built, deployed
Demo video call	[fill in: live call on the deployed build, or scripted Rep on the real console, labelled]
Telephony (Twilio SIP), live Mode B, payer inconsistency index	roadmap

Statement extraction is rule-based on the demo vocabulary; a differently phrased denial is captured as a verbatim quote, not as a typed fact. That boundary is deliberate: the model never touches evidence.

Market: the denial-chasing spend, not the phone call

\$25.7B

provider cost of claims
adjudication, up 23% year on
year

\$57.23

administrative cost per
denied claim

73%

of providers cite unclear
denial reasons or
underpayments

Target user: outsourced medical billing companies and hospital central billing offices. Named teams with a denial queue, not "healthcare".

Sizing: CAQH's published \$2.4B medical savings opportunity for claim-status transactions. We do not reconstruct a volume-times-price figure; only 2% of 2.82B inquiries are fully manual.

Revenue model

- **Per seat, per month** for billing teams, priced against the 25 minutes a call costs today.
- **Per-call metering** above the seat quota, so a large central billing office pays for what it uses.
- **The appeal packet** is the upsell: an evidence-grade record the payer must treat as a relevant document.

Why it needs AI: the record has to be made while the call is happening, by something that can hear a reference number, compare it to six earlier calls and speak in under two seconds. A person on the call cannot do that and take notes.

Competitive analysis

	Makes the call	Records it	Remembers across calls	Flags contradictions live	Citable evidence
AI payer-call agents (Infinitus, Outbound AI)	yes	yes	per task	no	no
Denial-management platforms (Waystar, Experian Health, AKASA)	no	no	claim status only	no	no
Call recording and QA tools	no	yes	no	no	audio only
Status quo: claim notes	the biller does	no	a line of shorthand	no	no
The Witness	rides along or speaks	yes	whole claim history	under 2 s, spoken	quotes, hash, offsets

Our wedge: cross-call payer memory and an evidence-grade record. We witness the call rather than replace the biller, so nothing about the workflow changes on day one.

Roadmap, and the line we will not cross

- Telephony: Twilio SIP into the same two sockets.
- Live Mode B: the biller speaks, the Witness whispers, on a second Streaming session.
- Payer Inconsistency Index: "this payer gave inconsistent answers on 3 of 11 claims this quarter".
- Audio snippets in the appeal packet beside every quote.

The console names the representative, because a statement without an attributable speaker is not citable.

We will not build rep scoring. Ranking call-centre workers by contradiction rate is a surveillance product wearing an evidence product's clothes. Attribute individuals because evidence requires it; aggregate only at the payer level.

Try it

<https://the-witness-omega.vercel.app>

<https://github.com/velez2689/the-witness>

Scripted path: press Start call. No microphone, no network, no session opens.

Live path: Be the Rep (live), headphones, Chrome. Two AssemblyAI sessions open on click and close on every exit path.

Next.js on Vercel. AssemblyAI Streaming STT v3 and Voice Agent API. MIT licence. All demo data synthetic; no PHI; no CPT descriptor text; no accuracy claims.

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